MIL-C-46635C(MU)
3 August 1973
SUPERSEDING
MIL-C-46635B
12 January 1967

MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH PRESSURE TEST, M54A1

1. SCOPE

1.1 This specification covers one type of electric primed cartridge for use in proof testing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258858 - Cartridge, 20mm, High Pressure Test, M54A1
IEL 7258858 - Index of Inspection Equipment Lists for: Cartridge, 20mm, High Pressure Test, M54A1
D7258928 - Marking on Shipping and Storage Container, M548, for 20mm Cartridges
F7258941 - Method of Packing 100 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258942 - Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container

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2.1 (Cont)

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing (Dwg.) D7258858, referenced specifications and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.4)

3.3 First article sample.-Unless otherwise specified, the supplier shall submit a preproduction sample.

3.4 Projectile extraction.-The tension necessary to separate the projectile from the cartridge case shall be as specified by Dwg. D7258858.

3.5 Electrical resistance, primed case.-The electrical resistance of the cartridge primer shall be not less than 500 ohms, or more than 1,200,000 ohms.

3.6 Projectile torque.-The projectile shall withstand a minimum torque as specified by Dwg. D7258858.

3.7 Pressure.-The cartridge, conditioned at 68°F to 72°F and fired, shall exhibit a chamber pressure between 62,500 pounds per square inch (psi) and 72,500 psi.

3.8 Workmanship.-The requirements for workmanship are as specified by applicable drawings, referenced specifications, and the following:

3.8.1 Process defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, cracks, splits, deep scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.8.2 Cleanliness.-The cartridge and its components shall be free of all foreign matter. (see 6.8)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First Article Sample.

4.2.1 Preproduction sample.-Prior to the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 125 cartridges. (see 6.3) The sample shall be manufactured using the same equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or other such facility specified in the contract. (see 6.3)

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Quality Conformance Inspection.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

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4.3.1.2 Lot identification.-Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a federal stock number assigned by the procuring agency.

4.3.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

4.3.2.1 Primed cartridge cases.-Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.4.2.1. Primed cases which fail to meet the requirements of 3.4 shall be rejected.

4.3.2.2 Sampling and acceptance criteria.-The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.3.2.2.1.c) In addition, where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects. (see 6.5)

4.3.2.2.1 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major	0.25%
Minor	0.40%

b. Class basis:

Major	1.50%
Minor	2.50%

c. Individual defects associated with the class basis:

Major	0.40%

4.3.3 Classification of defects.-1/ The classification of defects shall be as follows:

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4.3.3.1 Cartridge (see Dwg. D7258858).

Category and Defects	Method of Inspection
Critical:	
1. Split or perforated case	Visual
Major:	
101. Loose primer	Manual
102. Depth of primer seating	Gage
103. Over-all length, max.	Gage
104. Improper or missing crimp (case-projectile)	Visual
105. Profile and alignment, max.	Gage 2/
106. Corrosion	Visual
107. Head configuration	Visual
108. Metal slivers on head face	Visual
109. Length, case head to centerline crimp	Gage
Minor:	
201. Marking incorrect, incomplete, illegible or missing	Visual
202. Projectile protective coating damaged or incomplete	Visual
203. Waterproofing missing around primer	Visual
204. Scratch, dent, buckle, bulge, wrinkle or fold in case	Visual
205. Rotating band damaged	Visual
206. Primer crimp missing	Visual
207. Foreign matter, except corrosion	Visual
208. Incorrect type cartridge	Visual

1/ MIL-STD-651 shall apply in defining and evaluating visual defects.

2/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

4.3.4 Tests.-The tests listed in Table I shall be performed on each lot in accordance with the test methods prescribed in 4.4. For acceptance, each test shall comply with the applicable requirement(s). The sampling plans for the various tests shall be as specified in Table I.

Table I

Test		Sample Size Production Lot	Rqmt Para
Projectile extraction	1/	20	3.3
Electrical resistance verification	2/	20	3.4
Propellant contamination	3/	20	3.7.2
Projectile torque	4/	20	3.5
Pressure	5/	20	3.6

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4.3.4 (Cont)

1/ Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the minimum requirement. (see 3.3)

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to comply with the maximum requirement. (see 3.3)

2/ Electrical resistance verification.-The lot will be rejected if any primer of the sample cartridges fails to comply with the primer resistance requirements. (see 3.4)

3/ Propellant contamination.-The presence of foreign matter in the propellant of any of the test cartridges will cause rejection of the lot. (see 3.7.2 and 6.8)

4/ Projectile torque.-Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the accumulated samples fail to comply with the applicable requirement. (see 3.5)

5/ Pressure.-The lot shall be rejected if the sample fails to meet each of the following criteria: (see 6.6)

a. The sample average range ($\overline{R}$) shall not exceed 4,720 psi.

b. The sample average ($\overline{X}$) plus .95$\overline{R}$ shall not exceed the upper specified limit, i.e.

$$\overline{X} + .95\overline{R} \leq 72{,}500 \text{ psi.}$$

c. The sample average ($\overline{X}$) minus .95$\overline{R}$ shall equal or exceed the lower specified limit, i.e.

$$\overline{X} - .95\overline{R} \geq 62{,}500 \text{ psi.}$$

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4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.3.6 Inspection equipment.-Index of Inspection Equipment Lists (IEL) 7258858 identifies the applicable inspection equipment list required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to inspection equipment list for definition of coding, Dwg. B11075228, a detail of IEL 7258858).

4.4 Test methods and procedures.

4.4.1 Projectile extraction.-The method of test shall be as prescribed by AMCR 715-505, Volume 8.

4.4.2 Electrical resistance.

4.4.2.1 Primed cartridge cases.-The equipment and circuitry specified by IEL 7258858 shall be used to measure the resistance of the primer in each primed cartridge case.

4.4.2.2 Verification.-A Simpson Volt-Ohm-Ammeter, Model 260, or approved equal, will be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX 100 and RX 10,000 scales of the meter will be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum.

4.4.3 Propellant contamination.-The propellant from each of the cartridges tested for projectile extraction will be examined visually for contamination, as prescribed by AMCR 715-505, Volume 8.

4.4.4 Projectile torque.-Equipment shall be as specified by IEL 7258858. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque as specified by 3.5 shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, is detected by misalignment in the scratch mark.

4.4.5 Pressure.-The methods of test shall be as prescribed by AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed. (see 6.6)

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with Dwg. F7258941 or F7258942 as specified by the procuring agency.

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5.2 Marking.-Marking on the packed container shall be in accordance with Dwg. D7258928, as applicable.

6. NOTES

6.1 Intended use.-These cartridges are intended to be used only in proof testing 20mm guns and barrels chambered for the M103 cartridge case.

6.2 Ordering data.-Invitation for bids and contracts or orders will specify the following:

6.2.1 Title, number, and date of this specification.

6.2.2 Place of inspection, if not at place of manufacture.

6.2.3 First article sample requirements. (see 4.2)

6.2.4 Detailed packing and marking instructions. Dwg. F7258942 pack preferred. (see 5.1)

6.2.5 Coded markings required by Dwg. D7258928. (see 5.2)

6.2.6 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.2.7 Provision for submission and approval of the manufacturing process. (see 3.2)

6.3 First article sample.-In order to minimize inspection costs, the tests of 4.3.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements.

6.4 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.5 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.6 Computation, average range.-The individual values will be recorded in subgroups of five (5) in the order which they are observed. Each subgroup range (R) will be determined which is the arithmetic difference between the highest and lowest values observed in the subgroup. The ranges of the subgroups will be added and the sum of the ranges will be divided by the number of the subgroups to determine the average range ($\bar{R}$).

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6.7 International agreements.-Certain provisions of this specification are the subject of international standardization agreement STANAG 3585. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.8 Contamination of explosive components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfires, increase action time and lot rejection.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1305-A728

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-46635C(MU), Cartridge, 20MM, High Pressure Test, M54A1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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